

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
• Ship 10+ production AI features end-to-end: multi-model LLM pipelines (Claude, GPT-4o, Gemini), Temporal.io agentic workflows, and REST API integrations; 923 commits across 7 production deployments. • Own backend stack (PostgreSQL, Redis, Auth0 RBAC, Docker, Terraform, Sentry); debug and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
• Deploy cloud genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. • Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
• Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. • Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
• Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. • Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
• Master of Science in Computer Science • Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
• Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

KubeServe	Kubernetes LLM Inference Platform	2026
• Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving (Claude, GPT-4o, Gemini); circuit breaker + EWMA load balancing. • Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.		
LLM Eval Harness	Independent Project	2025
• Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, reasoning, and code generation tasks. • Tracked hallucination rate, p50/p95 latency, cost-per-call; applied bootstrap CIs and McNemar tests across 6 metrics.		
VectorLift	Semantic Search & Ranking Engine	2026
• Built production-grade semantic search platform: Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI search API with per-stage latency tracking across 6 pipeline configurations. • Fine-tuned bi-encoder (sentence-transformers, MS MARCO) and cross-encoder re-ranker (MiniLM); evaluated with NDCG@10, MRR@10, MAP and paired bootstrap significance testing.		

Languages and Technologies

• Languages: Python, TypeScript, JavaScript, Java, SQL, Bash, C • AI/ML: Claude/GPT-4o/Gemini APIs, MCP, LangChain, RAG, LLM evaluation, HuggingFace, PyTorch, prompt engineering • Backend & APIs: FastAPI, Node.js, NestJS, REST APIs, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch • Infrastructure: Kubernetes, Docker, Helm, ArgoCD, AWS, GCP, Terraform, GitHub Actions, OpenTelemetry, Prometheus		
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Additional Experience and Awards

• IEEE Published: Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees. • National Hackathon Winner: Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo. • Lead at Google Developers Student Club: Led 4+ tech events (hackathons, conferences) reaching 500+ students. • Technical Writing: Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog		
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